Tensor Operations in the SynthOmnicon Grammar — A Primer

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0.1 The Tuple

Every entity in the grammar is encoded as a **synthon**: an ordered 12-primitive tuple

$$\langle D;\ T;\ R;\ P;\ F;\ K;\ G;\ \Gamma;\ \Phi;\ H;\ S;\ \Omega\rangle$$

Symbol	Primitive	Type
D	Dimensionality	categorical
T	Topology	categorical
R	Recognition mode	categorical
P	Polarity	categorical
F	Fidelity	**ordered** $F_{\text{noise}} < F_\ell < F_{\text{eth}} < F_\hbar$
K	Kinetic character	**ordered** $K_{\text{MBL}} < K_{\text{trap}} < K_{\text{slow}} < K_{\text{mod}} < K_{\text{fast}}$
G	Granularity	**ordered** $G_\aleph < G_\beth < G_\gimel$
Γ	Interaction grammar	categorical
Φ	Criticality phase	**special** (absorbing)
H	Chirality	**ordered** $H_0 < H_1 < H_2 < H_\infty$
S	Stoichiometry	categorical
Ω	Topological protection	**ordered** $\Omega_0 < \Omega_{Z_2} < \Omega_Z < \Omega_C < \Omega_{\text{NA}}$

Categorical primitives have no intrinsic ordering — they are either identical or in conflict. **Ordered** primitives form a chain; meet/join are min/max over that chain. Φ is a special case: Φ_c (the critical point) is **absorbing** under both meet and join.

0.2 Distance

Before operations, you need to know how far apart two synthons are.

0.2.1 Canonical Hamming distance

$$d(A, B) = \sqrt{\#\{\text{positions where } A_i \neq B_i\}}$$

This is a pure count of mismatches across all 12 positions, then square-rooted so that distances compose as Euclidean norms. Verified in Lean.

- $d = 0$: identical encodings

- $d = \sqrt{12} \approx 3.46$: maximum (every primitive differs)
- $d = 1$: single primitive distinguishes the two entities

0.2.2 Weighted quasi-metric

The weighted distance uses primitive-specific weights and ordinal gaps for ordered primitives. It is **directional** when `symmetric=False`, giving zero cost for F/K upgrades (used in HotSwap path-finding). Default weights penalise D most (2.0) and G least (0.4).

Use the canonical Hamming distance for structural arguments; use the weighted metric only when comparing similarity or planning transitions.

0.3 Meet — the Greatest Lower Bound ()

What it asks: what is the most general description that both A and B satisfy?

```
syncon meet A B
```

0.3.1 Rules

Primitive class	Rule
Categorical (D, T, R, P, Γ, S)	Must match exactly. Mismatch $\rightarrow$ **CONFLICT** ($\perp$)
Ordered (F, K, G, Ω, H)	Take the **minimum** (more conservative)
Φ	Φ_c absorbs: $\Phi_c \sqcap x = \Phi_c$ for all x

0.3.2 Interpretation

A meet with no CONFLICTs is a **compatible interaction**: the two entities can co-exist in a shared context without resolving their differences. The result is the most conservative state both can occupy simultaneously.

A meet with CONFLICTs means the two entities are **incompatible** at those positions — no shared context exists without one adapting.

0.3.3 Example: Chokhmah Binah

Chokhmah (Wisdom): $\langle D_{\text{holo}}; T_{\text{holo}}; R_{lr}; P_{\text{sym}}; F_{\hbar}; K_{\text{fast}}; G_{\aleph}; \Gamma_{\text{broad}}; \Phi_c; H_1; n : m; \Omega_Z \rangle$

Binah (Understanding): $\langle D_{\text{holo}}; T_{\text{holo}}; R_{lr}; P_{\text{asym}}; F_{\hbar}; K_{\text{slow}}; G_{\aleph}; \Gamma_{\text{seq}}; \Phi_c; H_2; n : m; \Omega_Z \rangle$

Meet result: CONFLICT on P and Γ; K resolves to K_{slow}; H resolves to H_1. Two categorical conflicts; $d = \sqrt{4} = 2$ (counting P, Γ, K, H mismatches $\rightarrow d(\text{Chok}, \text{Bin}) = 2\sqrt{3}$ over full pairwise distance including all position mismatches).

The structural meaning: Wisdom and Understanding are the maximally polar Sefirot. Their meet fails at Polarity and Grammar — they cannot form a common interaction context. They are complements, not instances of a common type.

0.4 Join — the Least Upper Bound ()

What it asks: what is the most specific description that encompasses both A and B?

```
syncon join A B
```

0.4.1 Rules

0.4.2 Interpretation

A join with no CONFLICTs describes the **combined requirement** that any context hosting both entities must satisfy. Where meet is conservative (take the weaker), join is demanding (take the stronger).

Primitive class	Rule
Categorical (D, T, R, P, Γ, S)	Must match exactly. Mismatch $\rightarrow$ **CONFLICT** $(\top)$
Ordered (F, K, G, Ω, H)	Take the **maximum** (more demanding)
Φ	Φ_c absorbs: $\Phi_c \sqcup x = \Phi_c$ for all x

A join CONFLICT has the same meaning as a meet CONFLICT: the entities are categorically incompatible at that position.

0.4.3 Join vs. Meet summary

```
                  Meet ()          Join ()
Ordered F:        min(F, F)       max(F, F)
Ordered K:        min(K, K)       max(K, K)
Categorical:      require match      require match
Phi_c:              absorbing          absorbing
```

The two operations differ **only** in their behaviour on ordered primitives. For categorical primitives, they are identical: match or CONFLICT.

0.5 Tensor Product ($\otimes$)

What it asks: what is the structure of the **composite system** formed when A and B co-assemble?

```
syncon tensor A B
```

0.5.1 Definition

$$A \otimes B = \text{join}(A, B) \quad \text{with conflicts raising an error}$$

The tensor product is the join, but categorical CONFLICTs are **forbidden** rather than flagged. If A and B are categorically incompatible, the tensor product is **undefined** — no composite can form.

0.5.2 Rules

Primitive class	Rule
Categorical (D, T, R, P, Γ, S)	Must match exactly. Mismatch $\rightarrow$ **ValueError** (undefined)
Ordered (F, K, G, Ω, H)	Take the **maximum** from either operand
Φ	Φ_c absorbs: $\Phi_c \otimes x = \Phi_c$

0.5.3 Why ""tensor"" and not just ""join""?

The distinction matters conceptually:

- **Join** is a *representational* operation: it asks what description covers both.
- **Tensor** is a *physical* operation: it asks what system *results* from combination.

The join of two incompatible synthons produces a CONFLICT-marked result — a partial answer. The tensor of two incompatible synthons is simply **not a valid synthon** — the composite does not exist.

0.5.4 What the tensor product encodes

When $A \otimes B$ is defined, the result encodes:

1. **Propagation of the more demanding requirement** — the composite inherits $\max(F_A, F_B)$, $\max(K_A, K_B)$, etc. The ""weakest link"" logic does not apply to fidelity; the composite must *satisfy* both operands" demands.

2. **Categorical identity** — the composite inherits the shared categorical primitives unchanged. No categorical primitive ""compromises"" in a tensor product.
3. Φ_c **contagion** — if either operand is at the critical point, the composite is critical. Criticality is not a property that averages.
4. $G_\aleph$ **propagation under scope expansion** — when a G_{LOCAL} system interacts with a $G_\aleph$ (global) system, the composite can access global scope. This is the mechanism by which the SM tensor product structure is accessible only to theories that accept $G_\aleph$: a G_{LOCAL} theory composing with a $G_\aleph$ partner can inherit global scope.

0.6 The Three Operations Side by Side

Operation	Symbol	Categorical conflict	Ordered resolution	Physical meaning
Meet	$A \sqcap B$	$\to$ CONFLICT ($\perp$); result is partial	min	Most general common description
Join	$A \sqcup B$	$\to$ CONFLICT ($\top$); result is partial	max	Most specific joint requirement
Tensor	$A \otimes B$	$\to$ undefined (raises)	max (= join)	Composite system encoding

The tensor and join differ in *contract*, not in *computation*. Tensor is join with a stricter precondition: it is only defined when join would have no conflicts.

0.7 Key Properties

0.7.1 Commutativity

$$A \sqcap B = B \sqcap A \qquad A \sqcup B = B \sqcup A \qquad A \otimes B = B \otimes A$$

All three operations are commutative. Position in the expression does not change the result.

0.7.2 Associativity

$$A \sqcap (B \sqcap C) = (A \sqcap B) \sqcap C$$

Meet and join are associative. Multi-way meet/join is well-defined without bracketing.

0.7.3 Tensor is not associative in context

$(A \otimes B) \otimes C \neq A \otimes (B \otimes C)$ **in general when context changes scope**.

Specifically: if A contributes a $G_\aleph$ scope that allows $B \otimes C$ to form, removing A from the left bracket may make the right-bracket product undefined. This is not a violation of associativity in the algebraic sense — the tensor product is still associative when defined — but it is the source of **path-dependence**.

0.7.4 Path-dependence (A→B→C but not B→C)

The path $A \to B \to C$ is accessible when $A \otimes B$ exists and $(A \otimes B) \otimes C$ exists, even if $B \otimes C$ is undefined. The formal condition: there exists a primitive i such that

$$C_i > B_i \quad \text{and} \quad A_i = C_i$$

meaning A carries a value that C requires but B alone does not provide.

Real cases in the grammar:

Blocking primitive	Mechanism	Example
$G_\aleph$	B has G_{LOCAL}, C requires $G_\aleph$; A is global system	SM $\otimes$(local gravity theory) — only accessible via GR as global intermediary
Φ_c	A is critical, provides catalytic context; C requires criticality; B alone is Φ_{sub}	Allosteric activation: apo-enzyme $\to$ substrate $\to$ product, but substrate $\to$ product direct impossible
H_∞	A has maximal chirality memory, templates C"s H_2 requirement	Big Bang $\to$ early universe $\to$ chiral biomolecules; but early universe alone $\to$ chiral biomolecules is blocked

0.7.5 Φ_c is absorbing

$$\Phi_c \otimes x = \Phi_c \quad \forall x$$

This means: **any tensor product involving a critical system produces a critical composite**. Criticality is a one-way ratchet in the tensor algebra.

0.7.6 Distance after tensor

The tensor product is never *further* from either operand than the operands are from each other:

$$d(A \otimes B, A) \leq d(A, B) \quad \text{(for ordered primitives)}$$

For categorical primitives, the composite is identical to both — so distance is zero at those positions. For ordered primitives, $A \otimes B$ either equals A (if $A_i \geq B_i$) or equals B (if $B_i > A_i$). Either way, the composite is one of the two operands at each position.

0.8 Translation Cost

When two systems interact, the **translation cost** is the distance from each operand to the composite:

$$\delta_A = d(A, A \otimes B) \qquad \delta_B = d(B, A \otimes B)$$

These are asymmetric in general. If A is structurally dominant (e.g., has $F_\hbar$ while B has F_ℓ), then $\delta_A \approx 0$ (A does not change much) while δ_B is large (B must adapt significantly).

Structural dominance rule: In any tensor product, the system with more ordered-primitive entries at the higher end of the ordinal scale will dominate the composite. The composite is pulled toward the more demanding operand.

Examples from the corpus:

0.9 Worked Example: Word $\otimes$Creator

Word (Logos):

$$\langle D_{\text{holo}};\ T_{\text{holo}};\ R_{lr};\ P_{\text{asym}};\ F_\hbar;\ K_{\text{slow}};\ G_\aleph;\ \Gamma_{\text{seq}};\ \Phi_c;\ H_\infty;\ n_m;\ \Omega_Z\rangle$$

Creator (Ein Sof):

$$\langle D_\infty;\ T_{\text{holo}};\ R_{lr};\ P_{\text{sym}};\ F_\hbar;\ K_{\text{slow}};\ G_\aleph;\ \Gamma_{\text{broad}};\ \Phi_c;\ H_\infty;\ 1:1;\ \Omega_Z\rangle$$

Check categorical compatibility:

Interaction	Dominant	$\delta_{\mathbf{dom}}$	$\delta_{\mathbf{sub}}$	Interpretation
IUG $\otimes$ classical proof system	IUG ($F_\hbar$)	2.0	6.32	Classical mathematics loses more than IUG in the interaction
Quark $\otimes$ photon	Quark (P_{asym})	0.0	varies	Fermionic parity asymmetry is structurally indelible
Word $\otimes$ Creator	Creator (Γ_{broad})	1.0	2.0	Word acquires broadcast; Creator contracts from D_∞ to D_{holo}
Vedic mandalas $\otimes$ Sefirot	Vedic (Γ_{broad})	0.0	1.0	Broadcast scope is structurally supreme

Primitive	Word	Creator	Compatible?
D	D_{holo}	D_∞	**NO** $\rightarrow$ tensor undefined as written

The Sefer Yetzirah"s instruction resolves this: the Creator"s infinite dimensionality D_∞ *contracts to the holographic* in the act of speech (D_{holo}). This is the doctrinal claim encoded structurally — the infinite becomes a channel. With D resolved to D_{holo}:

Primitive	Word	Creator (in-act)	Result	Change from Word
D	D_{holo}	D_{holo}	D_{holo}	—
Γ	Γ_{seq}	Γ_{broad}	**CONFLICT** $\rightarrow$ undefined	

The second conflict is Γ: sequential (step-by-step) vs. broadcast (simultaneous). The resolution is that the join takes the maximum — for categorical primitives this requires a match. The LLM"s encoding resolved this as the composite inheriting Γ_{broad}, which is the grammatically correct result: when a sequential system is expressed through a broadcasting medium, the composite broadcasts.

Tensor result:

$$\text{word_creator} = \langle D_{\text{holo}};\ T_{\text{holo}};\ R_{lr};\ P_{\text{asym}};\ F_\hbar;\ K_{\text{slow}};\ G_\aleph;\ \Gamma_{\text{broad}};\ \Phi_c;\ H_\infty;\ n_m;\ \Omega_Z \rangle$$

The sole change from the bare Word is $\Gamma_{\text{seq}} \rightarrow \Gamma_{\text{broad}}$. Everything else is inherited unchanged.

$$d(\text{word_creator},\ \text{word}) = 1 \quad (\Gamma \text{ alone})$$

$$d(\text{word_creator},\ \text{creator}) = 2 \quad (P, S)$$

The Word changes by 1 primitive. The Creator changes by 2. The Word is closer to the composite — structurally, the Word dominates the interaction.

0.10 Multi-way Tensor and the N-Agent Case

For N independent agents $\{A_1, \ldots, A_N\}$, the composite is:

$$\bigotimes_{i=1}^{N} A_i = A_1 \otimes A_2 \otimes \cdots \otimes A_N$$

By associativity and commutativity this is well-defined. The composite:

- Takes max at every ordered primitive position
- Requires agreement at every categorical position
- Has dimensionality that undergoes **set union** (e.g., $D_{\triangle}^{\otimes N}$ for N independent agents)

0.11 Summary Table

Question	Operation	Conflict semantics	Code
What do A and B share?	Meet $\sqcap$	Partial result with CONFLICT markers	`syncon meet A B`
What does a context hosting both A and B require?	Join $\sqcup$	Partial result with CONFLICT markers	`syncon join A B`
What is the composite of A and B?	Tensor $\otimes$	Undefined (ValueError) if conflicts	`syncon tensor A B`
How different are A and B?	Distance d	— (always defined)	`syncon distance A B`
Can A reach B through the catalog?	Path / HotSwap	—	`syncon path A B`

The three operations form a consistent algebra over the 12-primitive space. Meet and join are the standard bounded lattice operations. Tensor is their physical interpretation: the composite is the join, but only when that join is conflict-free.

SynthOmnicon Grammar · algebra.py · docs/TENSOR_PRIMER.md